Phishing Incident Response and Alert Triage

1. Project Overview

This project documents a simulated phishing incident investigation based on a practical cybersecurity activity from Coursera.

In this scenario, I assumed the role of a Level 1 Security Operations Center (SOC) analyst at a financial services company. The organization received an alert about a suspicious email containing a potentially malicious attachment. The attachment's SHA-256 hash had already been investigated and confirmed as malicious in a previous course activity.

My task was to evaluate the alert, identify phishing indicators, follow the incident response playbook, document the findings, and decide whether to close or escalate the alert.

This exercise demonstrates my understanding of basic SOC workflows, phishing investigation, security alert triage, incident documentation, and evidence-based escalation.

2. Investigation Objectives

The main objectives of this project were to:

- Evaluate a phishing alert and its severity.
- Analyze suspicious sender information and email content.
- Identify potentially malicious email attachments.
- Understand the role of SHA-256 hashes in file identification and threat intelligence.
- Follow a structured phishing incident response playbook.
- Document an incident using the 5 Ws.
- Update an alert ticket with investigation findings.
- Make an appropriate escalation decision based on available evidence.

3. Incident Scenario

An employee at a financial services company received an email related to an engineering job application. The message included a résumé and cover letter as a password-protected attachment.

The attachment was named **bfsvc.exe**. The email provided a password to open the file, and the attachment had reportedly been downloaded and opened on the employee's computer.

The file's SHA-256 hash had previously been verified as malicious within the course scenario.

As the Level 1 SOC analyst, my responsibility was to assess the available evidence, follow the organization's phishing response playbook, and escalate the incident when the findings indicated a potential security threat.

4. Alert Ticket Details

Ticket ID: A-2703
Alert Source: SERVER-MAIL
Alert Message: Phishing attempt — possible download of malware
Alert Severity: Medium
Ticket Status: Escalated
Attachment Name: bfsvc.exe
Investigation Type: Phishing incident response

File Hash

SHA-256:

```
54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b
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Hash Investigation Result: The hash was previously identified as malicious in the course activity. This project relies on that finding and does not claim an independent malware analysis.

5. Email Evidence

The alert ticket provided the following email information:

Sender Display Name: Def Communications
Sender Email Address: 76tguy6hh6tgftrt7tg.su
Sender Signature Name: Clyde West
Recipient: hr@inergy.com
Date and Time: July 20, 2022, at 09:30:14 AM
Subject: Re: Infrastructure Egnieer role
Attachment: bfsvc.exe

The email claimed that the sender was applying for an engineering position and had attached a résumé and cover letter. It also stated that the attachment was password-protected and provided a password for opening it.

These details were evaluated as part of the phishing investigation.

6. Phishing Indicators Identified

6.1 Suspicious Sender Information

The sender's email address appeared unusual and inconsistent with the display name, "Def Communications," and the signature name, "Clyde West."

This inconsistency raised concerns about the sender's identity and the legitimacy of the email.

6.2 Grammatical Errors

The email subject contained a spelling error, and the message body included awkward phrasing and grammatical mistakes.

Although grammatical errors alone do not prove that an email is malicious, they can support a phishing assessment when combined with other suspicious indicators.

6.3 Password-Protected Executable Attachment

The email contained an attachment named **bfsvc.exe** and instructed the recipient to use a supplied password to open it.

An unexpected executable attachment presented as a résumé or cover letter is suspicious because executable files can contain or deliver malicious code.

6.4 Previously Confirmed Malicious Hash

The SHA-256 hash of the attachment had already been investigated and verified as malicious in the preceding course activity.

This was the strongest technical indicator supporting escalation because it provided an established malicious-file finding within the scenario.

6.5 Medium Alert Severity

The alert was classified as Medium severity. According to the activity's playbook guidance, a Medium or High severity alert can be an indication that escalation is required.

Combined with the malicious hash and the other phishing indicators, the severity supported further investigation.

7. Incident Analysis: The 5 Ws

Who was involved?

An employee received the suspicious email from a sender identifying as Def Communications and signing as Clyde West. The sender's actual identity was not established by the available evidence.

What happened?

A suspicious email delivered a password-protected executable attachment named bfsvc.exe. The scenario states that the file was downloaded and opened, and its hash had previously been confirmed as malicious.

When did the incident occur?

The email details recorded July 20, 2022, at 09:30:14 AM. This is the timestamp of the email, not necessarily the exact time the attachment was opened.

Where did the incident occur?

The scenario involved an employee's computer at a financial services company. The email was addressed to hr@inergy.com.

Why did it happen?

The email used a job application theme and presented the attachment as a résumé and cover letter to encourage the recipient to open it. The attacker's ultimate objective was not confirmed by the available evidence.

8. Incident Response Methodology

The investigation followed the general workflow described in the course's phishing incident response playbook.

Step 1: Access and Review the Alert

The existing alert ticket was reviewed to understand the alert message, severity, sender information, email content, and attachment details.

Step 2: Update the Ticket Status

The ticket status was initially set to Investigating to indicate that the alert was under review.

Step 3: Evaluate the Alert

The email and attachment were assessed for suspicious characteristics. The investigation identified sender inconsistencies, grammatical errors, a password-protected executable, and a previously confirmed malicious file hash.

Step 4: Determine Whether Escalation Was Required

The available evidence indicated a credible phishing incident involving a known malicious attachment. The alert was therefore considered appropriate for escalation rather than closure.

Step 5: Document the Findings

The ticket comments were updated with a summary of the incident, the identified phishing indicators, and the reasons for escalation.

Step 6: Escalate the Alert

The final ticket status was set to **Escalated**, referring the case to a Level 2 SOC analyst for further investigation and response.

9. Final Alert Ticket Comments

The following comments summarize the investigation and decision:

The alert detected a potential phishing incident in which an employee downloaded and opened a malicious attachment named bfsvc.exe. The file hash was previously investigated and confirmed to be malicious. The sender's email address appeared suspicious and inconsistent with the display name, "Def Communications," and the signature name, "Clyde West." The email also contained grammatical errors and a password-protected executable attachment presented as a résumé and cover letter. Considering these indicators and the Medium alert severity, I escalated the ticket to a Level 2 SOC analyst for further investigation and response.

10. Final Decision and Justification

Decision: ESCALATE

The alert was escalated for the following reasons:

1. The attachment's SHA-256 hash had already been confirmed as malicious in the course scenario.
2. The email contained multiple phishing indicators, including suspicious sender information, grammatical errors, and a password-protected executable attachment.
3. The scenario reported that the attachment was downloaded and opened, creating a need to investigate possible endpoint compromise.

Closing the alert would not be appropriate based on the available evidence. Escalation allows a higher-level analyst to investigate the potential impact and determine the necessary containment and remediation actions.

11. Recommended Next Steps

The following actions would be appropriate for the next stage of the investigation, subject to the organization's policies and authorization:

- Investigate the affected endpoint to determine whether the executable actually ran.

- Review relevant endpoint and security logs for suspicious activity.
- Determine whether other employees received the same email or attachment.
- Assess whether any additional malicious files, persistence mechanisms, or unauthorized activities are present.
- Follow the organization's containment procedures if compromise is confirmed or reasonably suspected.
- Preserve relevant evidence and document the investigation.
- Remove the malicious file and restore the affected system when appropriate under the organization's incident response procedures.

These are recommended follow-up actions. They are not claimed to have been performed during the course activity.

12. Skills Demonstrated

This project helped me develop and practice the following cybersecurity skills:

- Phishing alert triage
- Email security analysis
- Identification of suspicious attachments
- SHA-256 file-hash interpretation
- Security alert severity assessment
- Incident response playbook application
- Incident Handler's Journal documentation
- Alert ticket management
- Evidence-based escalation
- SOC Level 1 investigation fundamentals
- Security incident reporting

13. Tools and Learning Resources

Coursera: Detection and Response course activity.

Phishing Incident Response Playbook: Used as the reference for the investigation workflow.

Alert Ticket Template: Used to record findings and the final escalation decision.

SHA-256: Used in the course scenario to identify the previously investigated malicious attachment.

This was a guided, simulated investigation. The activity did not require direct access to a live SIEM platform, endpoint detection system, or corporate network. No independent malware execution or laboratory analysis is claimed.

14. Key Learning Outcomes

Through this practical exercise, I learned that effective incident response requires more than identifying a suspicious email. A SOC analyst must evaluate multiple indicators, review available evidence, follow established procedures, document findings clearly, and choose an appropriate response.

I also learned the importance of distinguishing confirmed evidence from assumptions. For example, the malicious file hash was confirmed within the scenario, but the extent of any endpoint compromise remained undetermined.

The activity strengthened my understanding of how a Level 1 SOC analyst supports incident response by identifying potential threats, maintaining accurate records, and escalating cases that require deeper investigation.

15. Conclusion

This project demonstrates my foundational understanding of phishing incident response and SOC alert triage. By evaluating the email, identifying suspicious indicators, using the established malicious-hash finding, documenting the 5 Ws, and escalating the alert, I applied a structured approach to a simulated security incident.

This work forms part of my ongoing cybersecurity learning journey as I build practical knowledge in threat detection, security operations, incident response, and defensive security.

Project Classification: Educational cybersecurity case study
Final Outcome: Alert escalated to Level 2 SOC analyst
Evidence Basis: Coursera-provided scenario and previously established course findings

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Disclaimer: This project is based on a simulated educational scenario. It does not represent an investigation of a real organization or a claim of professional incident response experience. All findings and conclusions are limited to the evidence provided in the course activity.